CS554 Midterm #1

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I. Question

Given the following error rates for a four feature problem, what will be the result of the following attribute subset selection procedures?

a) Stepwise forward selection
b) Stepwise backward elimination
c) Floating search

Error Rates: (F_1), 0.23; (F_2), 0.15; (F_3), 0.16; (F_4), 0.24; (F_1, F_2), 0.18; (F_1, F_3), 0.19; (F_1, F_4), 0.20; (F_2, F_3), 0.16; (F_2, F_4), 0.29; (F_3, F_4), 0.22; (F_1, F_2, F_3), 0.19; (F_1, F_2, F_4), 0.25; (F_1, F_3, F_4), 0.24; (F_2, F_3, F_4), 0.26; (F_1, F_2, F_3, F_4), 0.29.

II. Question

Given a dataset of N instances and K classes, and the class labels of all instances, implement stratified K-fold cross-validation algorithm.

III. Question

The following table shows the midterm and final exam grades obtained for students in a database course.

Midterm Exam	Final Exam
72	84
50	63
81	77
74	78
94	90

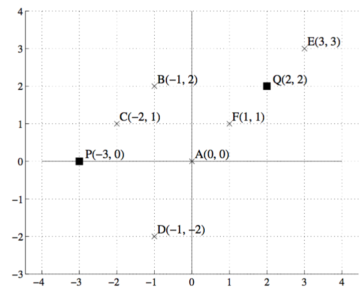
a) Derive a linear equation for the prediction of a student's final exam grade based on the student's midterm grade in the course.
b) Predict the final exam grade of a student who received an 86 on the midterm exam.

IV. Question

a) Suppose you train a classifier and test it on a held-out validation set. It gets 80% classification accuracy on the training set and 20% accuracy on the validation set. From what problem is your model most likely suffering? Underfitting, overfitting? Explain.
b) Suppose you train a classifier and test it on a held-out validation set. It gets 30% classification accuracy on the training set and 30% accuracy on the validation set. From what problem is your model most likely suffering? Underfitting, overfitting? Explain.

V. Question

Suppose we are given the following dataset. In this question, we will do k-means clustering to cluster the points A, B ... F into 2 clusters. The current cluster centers are P and Q.



a) Select all points that get assigned to the cluster with center at P.
b) What does cluster center P get updated to?
c) Suppose the new distance function is $d(A, B) = |A_x - B_x| + |A_y - B_y|$. Select all points that get assigned to the cluster with center at P.
d) What does cluster center P get updated to w.r.t. new distance function?

VI. Question

Give example covariance matrices and mean vectors for Qda, Lda, Naive Bayes and K-Means classifiers.

VII. Question

Assume you wish to use the K-nearest neighbors algorithm on this dataset and set aside the last two examples as a tuning set.Would you prefer K = 1 or K = 3?

A	B	C	D	Output
T	T	F	T	T
T	T	T	T	F
F	F	T	F	T
F	T	F	T	F
T	T	F	F	T
T	F	T	T	F

VIII. Question

Consider the geometric distribution, which has $P(X = k) = (1-\theta)^{k-1}\theta$. Assume in our training data X took on the values 4, 2, 7, and 9.

a) Write an expression for the log likelihood of the data as a function of the parameter θ. Remember, the likelihood is calculated by multiplying the likelihoods of the data points. Log likelihood is log of likelihood.
b) Take the derivative of the expression in part a) solve for θ and calculate the maximum likelihood estimate.